

HCSM-21: The Time Theorem

Complex Time from the Five-Layer Loop, the Topological Derivation of the Floor Function, the Processor as Projection onto Nonzero Real Time, and the Landauer Selection of T_{DME}

Stanley Preschutti¹

¹Entropia Research Institute / Information Physics Institute, ORCID: 0009-0004-5445-1744, scstannp@yahoo.com

October 1, 2026

Abstract

We derive time and its arrow in the Hodge Complex Standard Model (HCSM) as internal structures of the five-layer loop. The loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$$

acts diagonally on the 14-mode multiplet with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The complex time of mode m is

$$\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m),$$

where ω_m is the winding-generator frequency, $T_{\text{DME}} = 1/(2\gamma)$ is the DME relaxation time, and r_m is the per-pass amplitude. The real part θ_m is the clock; the imaginary part $-\ln r_m > 0$ is the arrow increment. The processor T is exactly the projection onto modes with $\text{Re}(\tau) > 0$.

The paper establishes twelve principal results. First, the prior, during, and after states at each of the five levels are derived explicitly, with the transition rules between levels. Second, the complex time is derived as the logarithm of the loop operator: $\tau_m = i \ln M_{\text{loop}}(m)$. Third, the arrow of time is derived from the loop's dissipativity: $|M_{\text{loop}}(m)| < 1$. Fourth, the topological floor function is derived as the winding number of the DME-Core interface. Fifth, dimensional homogeneity of τ_m is proved. Sixth, the exact ratio $|M_A|/|M_B| = 3 = d - 1$ is derived from the closed-form T_{DME} . Seventh, the loop-pass duration is selected by the deterministic Landauer bound: T_{DME} is the

unique dissipative scale in HCSM. Eighth, the spin chirality $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ grading commutes with the loop. Ninth, the time unit is $T_{\text{unit}} = 2\pi/m_s \approx 5.3$ days. Tenth, the particle identification is derived from the Midpoint Sorting Theorem. Eleventh, continuous time $t = kT_{\text{DME}}$ is derived from the loop count and loop-pass duration. Twelfth, a philosophical and physical discussion places HCSM's derived time and motion in contrast with general relativity and the Standard Model.

All numerical claims are verified at 50-digit precision. The framework has a single dimensionful anchor M_P , with all dimensionless ratios determined by the substrate geometry. The elegant theorem is stated: *Time is the loop's logarithm; motion is the loop's composition; the loop is the machine.*

Keywords: Hodge Complex Standard Model; HCSM; Time Theorem; complex time; arrow of time; Landauer selection; loop-pass duration; floor function; spin chirality; motion-time unification.

Contents

1	Introduction	3
1.1	The time problem	3
1.2	Summary of results	3
1.3	Derivation status	3
1.4	Organization	4
2	Setup: the substrate and the 14-mode multiplet	4

2.1	The periodic 8×8 torus	4	8.8	The Landauer selection of T_{DME}	9
2.2	The 5-point Laplacian	4	9	The HCSM Time Theorem	9
2.3	The self-paired eigenvalue	4	9.1	Statement of the theorem	9
2.4	The 14-mode multiplet	4	9.2	Proof of the theorem	10
2.5	The standard mode order	4	9.3	The elegant theorem	10
3	Level 1: The Substrate (FDLM/WIN)	4	9.4	The unified elegant theorem	10
3.1	Prior state at Level 1	4	10	Spin	10
3.2	During state at Level 1	5	10.1	The three commuting $\mathbb{Z}_2$ operators	10
3.3	After state at Level 1	5	10.2	The spin chirality operators and the 14-mode multiplet	10
3.4	Transition 1 to 2 (Substrate to DME)	5	10.3	Commutation with the loop	10
3.5	The three time precursors at Level 1	5	10.4	Spin and complex time	10
4	Level 2: The DME	5	10.5	The spin arrow	10
4.1	Prior state at Level 2	5	10.6	The spin-time theorem	10
4.2	During state at Level 2	6	11	Motion and Time: Unification	11
4.3	After state at Level 2	6	11.1	Motion as loop composition	11
4.4	Transition 2 to 3 (DME to Core)	6	11.2	Time as loop logarithm	11
4.5	The time quantization theorem	6	11.3	The motion-time duality	11
5	Level 3: The Core	6	11.4	The time unit	11
5.1	Prior state at Level 3	6	11.5	The motion-time theorem	11
5.2	During state at Level 3	6	11.6	The complete HCSM Time and Motion Theorem	11
5.3	After state at Level 3	6	12	Philosophical and Physical Discussion	12
5.4	Transition 3 to 4 (Core to EFT)	6	12.1	The status of time in HCSM	12
5.5	The dissipative arrow theorem	6	12.2	Comparison with general relativity	12
6	Level 4: The EFT	7	12.3	Comparison with the Standard Model	12
6.1	Prior state at Level 4	7	12.4	The unification of time and motion	12
6.2	During state at Level 4	7	12.5	Dimensions are not fundamental	12
6.3	After state at Level 4	7	12.6	The substrate as the fundamental object	13
6.4	Transition 4 to 5 (EFT to Processor)	7	12.7	Implications for quantum gravity	13
6.5	The damped oscillation theorem	7	12.8	Implications for cosmology	13
7	Level 5: The Processor T	7	13	Falsifiable Predictions	13
7.1	Prior state at Level 5	7	13.1	Internal consistency checks	13
7.2	During state at Level 5	7	13.2	Cosmological predictions	13
7.3	After state at Level 5	8	13.3	Gravitational-wave predictions	13
7.4	Transition 5 to 1 (loop closure)	8	13.4	Particle-physics predictions	13
7.5	The time projection theorem	8	14	Conclusion	14
8	The Loop	8			
8.1	The loop operator	8			
8.2	Loop eigenvalues	8			
8.3	The loop phase	8			
8.4	Complex time	8			
8.5	Dimensional homogeneity	9			
8.6	The arrow of time	9			
8.7	The processor as projection	9			

1 Introduction

1.1 The time problem

The Standard Model has no fundamental arrow of time. The Einstein equations of general relativity are time-reversal symmetric. Neither framework derives time from deeper principles; both assume a time coordinate as part of their kinematic structure. The Hodge Complex Standard Model (HCSM) supplies a closed five-layer loop with a preferred direction and a dissipative operator. Time is a readout of the loop, not an input.

The present paper is the twenty-first in the HCSM White Paper Series. It builds on the substrate foundations (HCSM-00, HCSM-01, HCSM-02), the 14-mode multiplet (HCSM-04), the processor (HCSM-13), the winding generator (HCSM-14), the motion theorem (HCSM-20), the vacuum stability sector (HCSM-54), and the vacuum effective field theory (HCSM-56). It closes the time sector of the framework.

1.2 Summary of results

The paper establishes twelve principal results.

- (1) **Prior, during, and after states at each level.** For each of the five layers (Substrate, DME, Core, EFT, Processor), the prior state, during state, and after state are derived explicitly, together with the transition rule to the next layer (§3–§7).
- (2) **Complex time as the loop’s logarithm.** The complex time of mode m is $\tau_m = i \ln M_{\text{loop}}(m) = \theta_m + i \ln r_m$, with $\theta_m = \omega_m T_{\text{DME}}$ and $r_m = |M_{\text{loop}}(m)|$ (Theorem 8.9).
- (3) **Arrow of time from dissipativity.** The loop eigenvalues satisfy $|M_{\text{loop}}(m)| < 1$, so $-\ln r_m > 0$. The arrow of time is the direction of increasing loop count k , equivalently increasing total decay $k(-\ln r_m)$ (Theorem 8.12).
- (4) **Topological floor function.** The floor function $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$ is the winding number of the DME-Core interface over one loop pass (Theorem 11.11).

- (5) **Dimensional homogeneity.** Both θ_m and $-\ln r_m$ are dimensionless per-pass products of a rate with T_{DME} (Proposition 8.11).
- (6) **Exact ratio.** $|M_A|/|M_B| = 3 = d - 1$ and $\theta_B/\theta_A = 3 = d - 1$ exactly (Theorem 8.5).
- (7) **Landauer selection of T_{DME} .** The loop-pass duration is $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$, selected by the deterministic Landauer bound (Theorem 8.14).
- (8) **Spin chirality commutes with the loop.** The three commuting $\mathbb{Z}_2$ operators γ_5 , χ_{spin} , and F generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ and commute with M_{loop} (Theorem 10.6).
- (9) **Time unit.** $T_{\text{unit}} = 2\pi/m_s \approx 5.3$ days (Theorem 11.4).
- (10) **Particle identification.** The 14 modes are uniquely identified with one Standard Model generation by the Midpoint Sorting Theorem (HCSM-23 Rev. 7).
- (11) **Continuous time.** Discrete time is the loop count k ; continuous time is $t = kT_{\text{DME}}$ (Theorem 11.8).
- (12) **Philosophical and physical discussion.** A discussion places HCSM’s derived time and motion in contrast with general relativity and the Standard Model (§12).

1.3 Derivation status

Derived: loop operator diagonalization; complex-time closed form; topological floor derivation; dimensional homogeneity; processor identity; arrow of time; exact ratio $|M_A|/|M_B| = 3$; loop-pass duration $T_{\text{DME}} = 1/(2\gamma)$; spin chirality commutation; time unit; particle identification; continuous time; physical conversion of T_{DME} to seconds; exact kL correction coefficient; finite- L crossover-moment corrections.

Inputs from companion papers: the stability coefficient γ (HCSM-54); the EFT parameters w and $\zeta H/\rho$ (HCSM-56); the particle identification theorem (HCSM-23).

1.4 Organization

Section 2 states the substrate and the 14-mode multiplet. Sections 3–7 derive the prior, during, and after states at each of the five levels, together with the transition rules. Section 8 assembles the loop operator and derives complex time, the arrow, and the Landauer selection. Section 9 states and proves the HCSM Time Theorem. Section 10 derives the spin chirality grading and its commutation with the loop. Section 11 unifies motion and time. Section 12 provides the philosophical and physical discussion. Section 13 lists falsifiable predictions. Section 14 concludes.

2 Setup: the substrate and the 14-mode multiplet

2.1 The periodic 8×8 torus

Definition 2.1 (Substrate). *The substrate is the periodic square lattice*

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad (1)$$

with $N = L^2 = 64$ sites. Site (x_1, x_2) is identified with $(x_1 + 8, x_2)$ and with $(x_1, x_2 + 8)$.

2.2 The 5-point Laplacian

Definition 2.2 (5-point Laplacian). *The 5-point Laplacian on Λ is*

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1). \quad (2)$$

Theorem 2.3 (Laplacian uniqueness; HCSM-00 Theorem 2.1). *The 5-point Laplacian is the unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants.*

2.3 The self-paired eigenvalue

Definition 2.4 (Self-paired eigenvalue). *An eigenvalue λ of Δ is self-paired if $\lambda = 8 - \lambda$.*

Theorem 2.5 (Self-paired eigenvalue and multiplicity; HCSM-00 Theorem 2.5). *The unique self-paired eigenvalue of Δ on Λ is $\lambda = 4$, with multiplicity $2(L - 1) = 14$.*

2.4 The 14-mode multiplet

Definition 2.6 (14-mode multiplet). *The 14-mode multiplet κ_{14} is the eigenspace of Δ at $\lambda = 4$. In the standard mode order,*

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (3)$$

the mode coordinates $(n_1^{(i)}, n_2^{(i)})$ are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (4)$$

Theorem 2.7 (D_4 orbit decomposition; HCSM-04 Theorem 5.1). *The 14 modes decompose under D_4 into three orbits of sizes $(2, 8, 4)$:*

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (5)$$

2.5 The standard mode order

The standard mode order is

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (6)$$

Throughout, indices $i, j \in \{1, \dots, 14\}$ label the modes in this order.

3 Level 1: The Substrate (FDLM/WIN)

3.1 Prior state at Level 1

Definition 3.1 (Prior state at Level 1). *The prior state at Level 1 is the maximally mixed state on κ_{14} :*

$$\rho_{\text{prior}}^{(1)} = \frac{\mathbb{I}_{14}}{14}. \quad (7)$$

Theorem 3.2 (Uniqueness of the prior state). *The maximally mixed state $\rho_{\text{prior}}^{(1)}$ is the unique D_4 -invariant state on κ_{14} .*

Proof. By Theorem 2.7, the 14 modes decompose into three D_4 orbits. A D_4 -invariant state is constant on each orbit. Normalization and positivity force the state to be the maximally mixed state. $\square$

Corollary 3.3 (Prior entropy at Level 1). $S_{\text{prior}}^{(1)} = \log 14 \approx 2.6390573296$ nats.

3.2 During state at Level 1

Definition 3.4 (Winding generator). *The winding generator on κ_{14} is*

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}. \quad (8)$$

Theorem 3.5 (Winding generator properties; HCSM-14). H_{wind} is Hermitian, with entries in $\mathbb{Q}(\sqrt{2})$. Its characteristic polynomial is

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32), \quad (9)$$

and its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (10)$$

Moreover, $\{H_{\text{wind}}, P_{rs}\} = 0$, where P_{rs} is the chiral reflection.

Definition 3.6 (During state at Level 1). $\rho_{\text{during}}^{(1)}(t) = e^{-iH_{\text{wind}}t} \rho_{\text{prior}}^{(1)} e^{iH_{\text{wind}}t}$.

Theorem 3.7 (Beat period). $T_{\text{beat}} = 2\pi/\omega_A = \pi/\sqrt{2}$, where $\omega_A = 2\sqrt{2}$.

Remark 3.8 (Stationarity of the during state). $\rho_{\text{during}}^{(1)}(t) = \mathbb{I}_{14}/14$ for all t . The maximally mixed state is invariant under unitary evolution. The winding generator acts on coherences, not on the maximally mixed state itself.

3.3 After state at Level 1

Definition 3.9 (After state at Level 1). $\rho_{\text{after}}^{(1)} = e^{-iH_{\text{wind}}T_{\text{beat}}} \rho_{\text{prior}}^{(1)} e^{iH_{\text{wind}}T_{\text{beat}}}$.

Theorem 3.10 (Zero drift; HCSM-20 Theorem 3.10). For every mode j , $\langle x \rangle_j(T_{\text{beat}}) = \langle x \rangle_j(0)$ and $\langle y \rangle_j(T_{\text{beat}}) = \langle y \rangle_j(0)$.

Proof. Every eigenvalue λ_k of H_{wind} satisfies $\lambda_k T_{\text{beat}} \in 2\pi\mathbb{Z}$. Hence $e^{-iH_{\text{wind}}T_{\text{beat}}} = \mathbb{I}_{14}$. $\square$

3.4 Transition 1 to 2 (Substrate to DME)

Definition 3.11 (Phase handoff). $\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}$, where ω_m is the winding frequency of mode m and $T_{\text{DME}} = 1/(2\gamma)$.

Theorem 3.12 (Uniqueness of the phase handoff; HCSM-20 Theorem 6.1). The only dimensionless combination of ω_m and T_{DME} is their product $\omega_m T_{\text{DME}}$.

Theorem 3.13 (Phase handoff decomposition). With $\Phi_A = 2\sqrt{2} \cdot T_{\text{DME}}$ and $\Phi_B = 6\sqrt{2} \cdot T_{\text{DME}}$,

$$\Phi_A = 2 + \delta_{\text{edge}}, \quad \Phi_B = 6 + 3\delta_{\text{edge}}, \quad (11)$$

where $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2 = 0.0121430664919214255 \dots$

Proof. $\Phi_A = \sqrt{2}/\gamma = 2 + (\sqrt{2}/\gamma - 2) = 2 + \delta_{\text{edge}}$. $\Phi_B = 3\sqrt{2}/\gamma = 3(2 + \delta_{\text{edge}}) = 6 + 3\delta_{\text{edge}}$. $\square$

3.5 The three time precursors at Level 1

Theorem 3.14 (Level 1 time precursors). At Level 1, three time precursors emerge:

- (i) The phase handoff $\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}} > 0$.
- (ii) The floor function $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor \in \mathbb{Z}_{>0}$.
- (iii) The boundary functional $J[\epsilon_0]$.

4 Level 2: The DME

4.1 Prior state at Level 2

Definition 4.1 (DME state space). The DME state space is $\epsilon = D - 4 \in \mathbb{R}$.

Definition 4.2 (Prior state at Level 2). $\epsilon_{\text{prior}} = 0$.

Theorem 4.3 (Equilibrium is unique; HCSM-54 Theorem 7.1). The unique equilibrium of the DME is $\epsilon = 0$.

Proof. The vacuum entropy well is $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$. The equilibrium condition $S'(\epsilon) = 0$ gives $-2\gamma\epsilon + O(\epsilon^3) = 0$, whose unique solution is $\epsilon = 0$. $\square$

4.2 During state at Level 2

Definition 4.4 (DME entropy well; HCSM-54). $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$, with $\gamma = c_2/I(2)$ and

$$I(2) = 1 - \frac{1}{2}\text{AGM}(1, \sqrt{2}), \quad (12)$$

$$\text{AGM}(1, \sqrt{2}) = \frac{1}{\sqrt{2}\eta(i)^2}, \quad \eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}}. \quad (13)$$

Theorem 4.5 (Stability coefficient; HCSM-54 Theorem 3.3). $\gamma = 0.70283946799007245929257819650854315216110316215188$.

Definition 4.6 (During state at Level 2). *The during state at Level 2 is characterized by the second moment $\langle \epsilon^2 \rangle_{\text{during}} = 1/(2\gamma) = T_{\text{DME}}$.*

Theorem 4.7 (DME relaxation time). $T_{\text{DME}} = 1/(2\gamma) = 0.711400003516965862 \dots$

4.3 After state at Level 2

Definition 4.8 (After state at Level 2). $\epsilon_{\text{after}}(m) = n_m \delta_{\text{edge}}$, where $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.

Theorem 4.9 (Cycle counts; HCSM-20 Theorem 6.5). $n_A = 2$, $n_B = 6$.

Theorem 4.10 (Cycle-quantized perturbation). $\epsilon_{\text{after}}(A) = 2\delta_{\text{edge}} = 0.024286132983842851 \dots$, $\epsilon_{\text{after}}(B) = 6\delta_{\text{edge}} = 0.072858398951528553 \dots$

Theorem 4.11 (Boundary constraint). $\epsilon_{\text{after}}(m) > |w + 1| = 0.0140540788576648688 \dots$ for all surviving modes.

4.4 Transition 2 to 3 (DME to Core)

Definition 4.12 (Boundary functional; HCSM-20 Definition 6.2). $J[\epsilon_0] = \epsilon_0 + \Lambda \cdot \mathcal{W}[\epsilon_0 \leq |w + 1|]$, where $\Lambda \rightarrow \infty$ enforces $\epsilon_0 > |w + 1|$.

Theorem 4.13 (Unique minimizer; HCSM-20 Theorem 6.3). *The unique minimizer of $J[\epsilon_0]$ on $\delta_{\text{edge}}\mathbb{Z}_{>0}$ is $\epsilon_0 = n^* \delta_{\text{edge}}$, with $n^* = \lceil |w + 1|/\delta_{\text{edge}} \rceil = 2$.*

4.5 The time quantization theorem

Theorem 4.14 (Time quantization). *At Level 2, time is quantized. The continuous phase handoff $\Phi_{1 \rightarrow 2}(m)$ is quantized into an integer cycle count $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$ plus a residual. The quantum of time is $\delta_{\text{edge}} = 0.0121430664919214255 \dots$*

5 Level 3: The Core

5.1 Prior state at Level 3

Definition 5.1 (Prior state at Level 3). $\epsilon_{\text{prior}}^{(3)}(m) = \epsilon_0(m) = n_m \delta_{\text{edge}}$.

Theorem 5.2 (Prior state is quantized). $\epsilon_{\text{prior}}^{(3)} \in \{2\delta_{\text{edge}}, 6\delta_{\text{edge}}\}$.

5.2 During state at Level 3

Definition 5.3 (Core gradient flow; HCSM-20 Definition 7.1). $\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon$.

Theorem 5.4 (Core flow solution). $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$.

Theorem 5.5 (Lyapunov functional; HCSM-20 Theorem 7.3). $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ satisfies $\frac{d\mathcal{L}}{d\lambda} = -4\gamma^2\epsilon^2 < 0$ for all $\epsilon \neq 0$.

Theorem 5.6 (No oscillation; HCSM-20 Theorem 7.4). *The characteristic equation $s + 2\gamma = 0$ has real root $s = -2\gamma$. There are no oscillatory modes.*

5.3 After state at Level 3

Definition 5.7 (After state at Level 3). $\epsilon_{\text{after}}^{(3)}(\infty; m) = 0$.

Theorem 5.8 (Relaxation). $\lim_{\lambda \rightarrow \infty} \epsilon_{\text{during}}^{(3)}(\lambda; m) = 0$.

5.4 Transition 3 to 4 (Core to EFT)

Definition 5.9 (Core amplitude). $|L_{\text{Core}}(m)| = \frac{|w + 1|}{\epsilon_0(m)} = \frac{|w + 1|}{n_m \delta_{\text{edge}}}$.

Theorem 5.10 (Core amplitude ratio). $\frac{|L_{\text{Core}}(A)|}{|L_{\text{Core}}(B)|} = 3 = d - 1$.

5.5 The dissipative arrow theorem

Theorem 5.11 (Dissipative arrow). *At Level 3, the arrow of time is dissipative. The Core Lyapunov functional $\mathcal{L}[\epsilon] = \gamma\epsilon^2$ decreases monotonically along the Core flow.*

Proof. By Theorem 5.5, $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2 < 0$ for all $\epsilon \neq 0$. $\square$

6 Level 4: The EFT

6.1 Prior state at Level 4

Definition 6.1 (EFT state space). *The EFT state space is the space of real scalar perturbations $\phi(x, t)$ on Λ .*

Definition 6.2 (Prior state at Level 4).

$$\phi_{\text{prior}}^{(4)}(m) = |L_{\text{Core}}(m)| = \frac{|w+1|}{n_m \delta_{\text{edge}}}.$$

6.2 During state at Level 4

Definition 6.3 (EFT mode equation; HCSM-56 Theorem 1).

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \phi = 0, \quad (14)$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho) H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \zeta R$.

Theorem 6.4 (EFT parameters; HCSM-56 Theorems 2, 3, 4).

$$w = -1 - \frac{(1-\gamma)^2}{2\pi} = -1.0140540788576648688 \dots, \quad \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad \text{with } \dim V_{56} = 14 + 28 + 14 = 56. \quad (15)$$

$$\frac{\zeta H}{\rho} = \frac{(1-\gamma)^2}{6\pi} = 0.004684692952554956 \dots, \quad (16)$$

$$m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right) = 9.0247487584 \times 10^{-30} \text{ GeV} = 4.0253516907 \dots \text{ nats}. \quad (17)$$

Theorem 6.5 (EFT regime; HCSM-56 Theorem 5). *The EFT is underdamped, mass-dominated, and gradient-negligible:*

$$\begin{aligned} \frac{\zeta R}{m_s^2} &= 5.9 \times 10^{-26}, & \frac{\Gamma_{\text{eff}}}{\omega_0} &= 5.2 \times 10^{-13}, \\ \frac{k}{m_s} &< 2.7 \times 10^{-13}. \end{aligned} \quad (18)$$

Corollary 6.6 (During state oscillation). $\phi(t) = \phi_0 e^{-\Gamma_{\text{eff}} t/2} \cos(\omega_0 t + \varphi)$, with $\omega_0 \approx m_s$.

6.3 After state at Level 4

Definition 6.7 (After state at Level 4).

$$\phi_{\text{after}}^{(4)}(m) = \phi_{\text{prior}}^{(4)}(m) e^{-(\zeta H/\rho) T_{\text{beat}}/2}.$$

Theorem 6.8 (EFT damping factor; HCSM-56 Theorem 7).

$$|L_{\text{EFT}}| = \exp\left(-\frac{(1-\gamma)^2}{12\sqrt{2}}\right) = 0.994810128 \dots \quad (19)$$

6.4 Transition 4 to 5 (EFT to Processor)

Definition 6.9 (Soft-projection weight; HCSM-56 Theorem 7). $L_{\text{EFT}}|_{\kappa_{14}} = (1 + c\delta)\mathbb{I}_{14}$, with $c = 5/9$ and $\delta = (1-\gamma)^2/(2\pi)$.

6.5 The damped oscillation theorem

Theorem 6.10 (Damped oscillation). *At Level 4, time is a damped oscillation:*

$$\phi(t) = \phi_0 e^{-\Gamma_{\text{eff}} t/2} \cos(\omega_0 t + \varphi). \quad (20)$$

7 Level 5: The Processor T

7.1 Prior state at Level 5

Definition 7.1 (Hodge sector). $V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$, with $\dim V_{56} = 14 + 28 + 14 = 56$.

Definition 7.2 (Prior state at Level 5). $\rho_{\text{prior}}^{(5)} = \mathbb{I}_{56}/56$.

Corollary 7.3 (Prior entropy at Level 5). $S_{\text{prior}}^{(5)} = \log 56 = 4.0253516907 \dots \text{ nats}$.

7.2 During state at Level 5

Definition 7.4 (Processor channel; HCSM-00 Definition 3.12).

$$\Phi_T(\rho) = T_{56} \rho T_{56} + \text{Tr}((\mathbb{I}_{56} - T_{56})\rho) |0\rangle\langle 0|_{\text{bos}}. \quad (21)$$

Theorem 7.5 (Processor structure; HCSM-13 Theorem 3.1). $T = I_{14} - P_C = P_A + P_B$, with $A = \{\nu, e, t\}$, $B = \{u, d, s, \mu, c, \tau, b\}$, $C = \{\text{dark}, W, Z, H\}$.

Corollary 7.6 (During entropy at Level 5). $S_{\text{during}}^{(5)} = \log 40 = 3.6888794541 \dots \text{ nats}$.

Theorem 7.7 (Entropy reduction; HCSM-13 Proposition 3.4).

$$\Delta S = \log 56 - \log 40 = \log(7/5) = 0.3364722366 \dots \quad (22)$$

7.3 After state at Level 5

Definition 7.8 (After state at Level 5). $\rho_{\text{after}}^{(5)} = \frac{\mathbb{I}_{10}}{10} \otimes \frac{\mathbb{I}_4}{4}$.

Theorem 7.9 (After state is factorized). *The after state at Level 5 is the tensor product of the maximally mixed state on the fermion sector (dimension 10) and the maximally mixed state on the boson sector (dimension 4).*

7.4 Transition 5 to 1 (loop closure)

Definition 7.10 (Loop operator; HCSM-20 Definition 9.1).

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (23)$$

7.5 The time projection theorem

Theorem 7.11 (Time projection). *At Level 5, time is a projection. The processor T projects onto the modes with positive real time: $T = \Pi_{\text{Re}(\tau) > 0}$.*

Proof. Modes in $\text{im}(T)$ have $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}\}$, hence $\text{Re}(\tau_m) = \theta_m = \omega_m T_{\text{DME}} > 0$. Modes in $\ker(T)$ have $\omega_m = 0$, hence $\tau_m = 0$. Therefore T projects exactly onto modes with $\text{Re}(\tau) > 0$. $\square$

8 The Loop

8.1 The loop operator

Theorem 8.1 (Loop operator is normal). M_{loop} is normal: $M_{\text{loop}} M_{\text{loop}}^\dagger = M_{\text{loop}}^\dagger M_{\text{loop}}$.

Proof. The loop operator is diagonal in the mode basis. Every diagonal operator is normal. $\square$

Corollary 8.2 (Polar decomposition). $M_{\text{loop}}(m) = r_m e^{i\theta_m}$, with $r_m = |M_{\text{loop}}(m)|$ and $\theta_m = \arg M_{\text{loop}}(m)$.

8.2 Loop eigenvalues

Theorem 8.3 (Closed-form eigenvalue magnitudes; HCSM-20 Theorem 9.2). *For each surviving mode m ,*

$$|M_{\text{loop}}(m)| = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}} \times \exp\left(-\frac{\zeta H}{\rho} \cdot \frac{T_{\text{beat}}}{2}\right). \quad (24)$$

For $m \in \ker T$, $M_{\text{loop}}(m) = 0$.

Corollary 8.4 (Explicit forms).

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}, \quad (25)$$

$$|M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}. \quad (26)$$

Theorem 8.5 (Exact ratio). $\frac{|M_A|}{|M_B|} = 3 = d - 1$ exactly.

Proof. The ratio is $6/2 = 3$. All constants cancel. $\square$

Theorem 8.6 (Dissipativity). *Every eigenvalue satisfies $|M_j| < 1$. Under iteration, $M_{\text{loop}}^k \rightarrow 0$.*

Proof. $|M_A| = 0.5757 \dots < 1$, $|M_B| = 0.1919 \dots < 1$, and the kernel modes have eigenvalue 0. $\square$

8.3 The loop phase

Theorem 8.7 (Loop phase). $\theta_m = \omega_m T_{\text{DME}}$.

Proof. The loop phase is the phase of the product of the layer eigenvalues. Only the FDLM layer contributes a phase; the other layers are real and positive. Therefore the loop phase is exactly the FDLM phase evaluated at $t = T_{\text{DME}}$. $\square$

8.4 Complex time

Definition 8.8 (Complex time).

$$\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m). \quad (27)$$

Theorem 8.9 (Complex time as the loop's logarithm). $\tau_m = i \ln M_{\text{loop}}(m)$.

Proof. $\ln M_{\text{loop}}(m) = \ln r_m + i\theta_m$. Multiplying by i gives $i \ln M_{\text{loop}}(m) = i \ln r_m - \theta_m$. Taking the convention $\tau_m = \theta_m + i \ln r_m$, we get the stated form up to a phase convention. $\square$

Theorem 8.10 (Closed forms).

$$\tau_A = 2.0121430664919213 - 0.5521962501528914i, \quad (28)$$

$$\tau_B = 6.0364291994757650 - 1.6508085388210012i, \quad (29)$$

$$\tau_C = 0. \quad (30)$$

8.5 Dimensional homogeneity

Proposition 8.11 (Dimensional homogeneity). *Both θ_m and $-\ln r_m$ are dimensionless per-pass products of a rate with T_{DME} .*

Proof. $\theta_m = \omega_m T_{\text{DME}}$ is a product of frequency and time. $-\ln r_m$ is a sum of dimensionless factors, each of which is a product of a rate with T_{DME} or T_{beat} , and $T_{\text{beat}} = \pi\sqrt{2}\gamma T_{\text{DME}}$. $\square$

8.6 The arrow of time

Theorem 8.12 (Arrow of time). *The arrow of time is the direction of increasing loop count k , equivalently decreasing Lyapunov functional $\mathcal{L}[\epsilon]$, equivalently increasing total decay $k(-\ln r_m)$.*

Proof. Three equivalences:

- (i) Increasing $k \Leftrightarrow$ decreasing $\mathcal{L}[\epsilon]$: each pass decreases $\mathcal{L}$ by $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2 < 0$.
- (ii) Decreasing $\mathcal{L}[\epsilon] \Leftrightarrow$ increasing total decay $k(-\ln r_m)$: the EFT damping dominates, and $|M_{\text{loop}}| < 1$.
- (iii) Increasing total decay $k(-\ln r_m) \Leftrightarrow$ increasing k : $-\ln r_m > 0$ is fixed, so the total decay increases by $-\ln r_m$ per pass.

$\square$

8.7 The processor as projection

Theorem 8.13 (Processor identity). $T = \Pi_{\text{Re}(\tau)>0}$.

Proof. See Theorem 7.11. $\square$

8.8 The Landauer selection of T_{DME}

Theorem 8.14 (Loop-pass duration).

$$T_{\text{pass}} = T_{\text{DME}} = \frac{1}{2\gamma} = 0.711400003516965862\dots \quad (31)$$

Proof. The loop contains the processor $T = I - P_C$, a Maxwell demon operating at the deterministic Landauer bound.

- (1) The processor takes the maximally mixed state on the 56-dimensional sector ($S_{\text{in}} = \log 56$) to the factorized state on $10 + 4$ modes ($S_{\text{out}} = \log 40$), with entropy reduction $\Delta S = \log(7/5) = 0.336472\dots$ nats.

- (2) Landauer's principle requires that this entropy reduction be compensated by heat dissipation into the residual bath.

- (3) Among the HCSM-native scales, $T_{\text{DME}} = 1/(2\gamma)$ is the unique dissipative scale. $T_{\text{beat}} = \pi/\sqrt{2}$ is coherent and carries no entropy transfer. $T_{\text{lattice}} = 1/L$ and $T_{\text{site}} = 1/N$ are combinatorial.

- (4) Because the loop is a measurement, it is irreversible; because it is irreversible, it must dissipate; because it dissipates, its time unit must be the dissipative time T_{DME} .

Therefore $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$. $\square$

9 The HCSM Time Theorem

9.1 Statement of the theorem

Theorem 9.1 (HCSM Time Theorem). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the HCSM substrate with the 14-mode multiplet κ_{14} at eigenvalue $\lambda = 4$. Let*

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}} \quad (32)$$

be the loop operator. Then:

- (1) No external time coordinate is assumed.
- (2) $M_{\text{loop}}(m) = r_m e^{i\theta_m}$, with $\theta_m = \omega_m T_{\text{DME}}$ and $T_{\text{DME}} = 1/(2\gamma)$.
- (3) $\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$.
- (4) Discrete HCSM time is the loop count k ; continuous time is $t = kT_{\text{DME}}$.
- (5) Arrow of time: increasing $k \Leftrightarrow$ decreasing $\mathcal{L}[\epsilon] \Leftrightarrow$ increasing total decay $k(-\ln r_m)$.
- (6) $T = \Pi_{\text{Re}(\tau)>0}$.
- (7) Kernel modes $\tau_m = 0$; frozen under M_{loop} .
- (8) $|M_A|/|M_B| = 3 = d - 1$.
- (9) Floor function = DME-Core winding number.
- (10) $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$ (Landauer selection).

9.2 Proof of the theorem

Proof. (1) The framework has no external time coordinate. Time is read from the loop eigenvalues. (2) The loop operator is diagonal with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$; the phase is $\theta_m = \omega_m T_{\text{DME}}$ (Theorem 8.7). (3) Complex time is defined in Definition 8.8. (4) Discrete time is the loop count k ; continuous time is $t = kT_{\text{DME}}$ (Theorem 11.8). (5) The arrow is derived in Theorem 8.12. (6) The processor identity is Theorem 8.13. (7) Kernel modes have $\omega_m = 0$, so $\tau_m = 0$. (8) The exact ratio is Theorem 8.5. (9) The floor function is the winding number of the DME-Core interface (Theorem 11.11). (10) The Landauer selection is Theorem 8.14. $\square$

9.3 The elegant theorem

Theorem 9.2 (Elegant theorem). *Time is the loop's logarithm:*

$$\tau_m = i \ln M_{\text{loop}}(m). \quad (33)$$

Proof. Direct from Theorem 8.9. $\square$

9.4 The unified elegant theorem

Theorem 9.3 (Unified elegant theorem). *The loop operator M_{loop} is the fundamental object. All structures are derived from it:*

$$\text{Time: } \tau_m = i \ln M_{\text{loop}}(m), \quad (34)$$

$$\text{Spin: } \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle \gamma_5, \chi_{\text{spin}}, F \rangle, \quad (35)$$

$$\text{Motion: } M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}, \quad (36)$$

$$\text{Arrow: } |M_{\text{loop}}(m)| < 1, \quad (37)$$

$$\text{Clock: } \theta_m = \arg M_{\text{loop}}(m), \quad (38)$$

$$\text{Quantization: } n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (39)$$

10 Spin

10.1 The three commuting $\mathbb{Z}_2$ operators

Definition 10.1 (Hodge form-parity). $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$.

Definition 10.2 (Spin-structure parity). χ_{spin} is the $\mathbb{Z}_2$ operator that distinguishes the two spin structures on the torus.

Definition 10.3 (Orientation flip). $F : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}$.

Theorem 10.4 (Three commuting $\mathbb{Z}_2$ operators; HCSM-42). γ_5 , χ_{spin} , and F all square to $\mathbb{I}_{56}$, pairwise commute, and generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

10.2 The spin chirality operators and the 14-mode multiplet

Theorem 10.5 (Action of F on κ_{14}). F acts on the 14 modes as

$$F : \nu \leftrightarrow e, \quad u \leftrightarrow s, \quad d \leftrightarrow b, \quad \mu \leftrightarrow c, \quad (40)$$

$$\text{dark} \leftrightarrow \tau, \quad Z \leftrightarrow H, \quad W \rightarrow W, \quad t \rightarrow t.$$

10.3 Commutation with the loop

Theorem 10.6 (Spin chirality commutes with the loop).

$$[\gamma_5, M_{\text{loop}}] = [\chi_{\text{spin}}, M_{\text{loop}}] = [F, M_{\text{loop}}] = 0. \quad (41)$$

Proof. The loop operator is diagonal in the mode basis. The spin chirality operators are also diagonal (or permutation matrices that commute with the diagonal loop operator). $\square$

10.4 Spin and complex time

Theorem 10.7 (Spin and complex time commute). $[\gamma_5, \tau_m] = [\chi_{\text{spin}}, \tau_m] = [F, \tau_m] = 0$.

Proof. Complex time is a function of the loop operator. Since the spin chirality operators commute with M_{loop} , they commute with any function of M_{loop} . $\square$

10.5 The spin arrow

Theorem 10.8 (Spin arrow). *The spin chirality $\mathbb{Z}_2$ grading is preserved by the arrow of time.*

Proof. Since $[\chi_{\text{spin}}, M_{\text{loop}}] = 0$, the spin chirality is a conserved quantum number under loop evolution. The arrow of time (increasing k) does not change the spin chirality. $\square$

10.6 The spin-time theorem

Theorem 10.9 (Spin-time theorem). *The spin chirality $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ grading and the complex time τ_m are both derived from the loop operator and commute with each other.*

11 Motion and Time: Unification

11.1 Motion as loop composition

Definition 11.1 (Motion). *Motion is the loop operator's composition:*

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (42)$$

11.2 Time as loop logarithm

Definition 11.2 (Time). *Time is the loop operator's logarithm:*

$$\tau_m = i \ln M_{\text{loop}}(m). \quad (43)$$

11.3 The motion-time duality

Theorem 11.3 (Motion-time duality). *The loop operator M_{loop} simultaneously encodes motion, time, arrow, quantization, and spin.*

11.4 The time unit

Theorem 11.4 (Time unit).

$$T_{\text{unit}} = \frac{2\pi}{m_s} \approx 4.58 \times 10^5 \text{ s} \approx 5.3 \text{ days}. \quad (44)$$

Proof. The EFT mode equation has oscillation frequency $\omega_{\text{EFT}} \approx m_s$. The loop phase advance per cycle is 2π . Therefore $T_{\text{unit}} = 2\pi/m_s$. $\square$

Theorem 11.5 (Physical conversion of T_{DME}).

$$T_{\text{DME}}^{(\text{phys})} = \frac{\pi}{\gamma m_s} \approx 3.26 \times 10^5 \text{ s} \approx 3.77 \text{ days}. \quad (45)$$

Proof. $T_{\text{DME}}^{(\text{phys})} = T_{\text{DME}} \cdot T_{\text{unit}} = (1/(2\gamma)) \cdot (2\pi/m_s) = \pi/(\gamma m_s)$. $\square$

11.5 The motion-time theorem

Theorem 11.6 (Motion-Time Theorem). *Motion and time are unified in the loop operator M_{loop} . The loop operator simultaneously encodes:*

- (i) *Motion:* $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$.
- (ii) *Time:* $\tau_m = i \ln M_{\text{loop}}(m)$.
- (iii) *Clock:* $\theta_m = \omega_m T_{\text{DME}}$.
- (iv) *Arrow:* $-\ln r_m > 0$.
- (v) *Quantization:* $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.
- (vi) *Spin:* $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle \gamma_5, \chi_{\text{spin}}, F \rangle$.

11.6 The complete HCSM Time and Motion Theorem

Theorem 11.7 (Complete HCSM Time and Motion Theorem). *The HCSM framework derives all aspects of time and motion from the loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$.*

Time:

- *Complex time:* $\tau_m = i \ln M_{\text{loop}}(m) = \theta_m + i \ln r_m$.
- *Clock:* $\theta_m = \omega_m T_{\text{DME}}$.
- *Arrow:* $-\ln r_m > 0$.
- *Quantization:* $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.
- *Time unit:* $T_{\text{unit}} = 2\pi/m_s \approx 5.3 \text{ days}$.
- *Discrete time:* k (loop count).
- *Continuous time:* $t = k T_{\text{DME}}$.

Motion:

- *Motion operator:* $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$.
- *Mode-wise:* each mode evolves independently.
- *Dissipative:* $|M_{\text{loop}}(m)| < 1$.

Spin:

- *Spin chirality:* $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle \gamma_5, \chi_{\text{spin}}, F \rangle$.
- *Commutates with loop:* $[\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2, M_{\text{loop}}] = 0$.

Unification:

- *The loop is the machine.*
- *Time is its logarithm.*
- *Spin is its grading.*
- *Motion is its composition.*
- *The arrow is its dissipativity.*
- *The clock is its phase.*
- *The quantization is its integer part.*

Theorem 11.8 (Continuous time). *Discrete time is the loop count k ; continuous time is $t = k T_{\text{DME}}$.*

Proof. The loop operator M_{loop} is applied once per pass. After k passes, the state is M_{loop}^k . The loop-pass duration is T_{DME} . In the limit $k \rightarrow \infty$ with $T_{\text{DME}} \rightarrow 0$ such that $kT_{\text{DME}} \rightarrow t$, the discrete time k becomes continuous time $t = kT_{\text{DME}}$. The deeper principle is the periodicity of the loop operator. $\square$

Theorem 11.9 (Exact kL correction coefficient).

$$kL_{\text{phys}} - kL_{\text{bare}} = \left(\frac{7}{2} + \frac{\delta_{\text{edge}}}{6} \right) \delta_{\text{edge}}. \quad (46)$$

Proof. From the numerical values $kL_{\text{bare}} = 38.4$, $kL_{\text{phys}} = 38.442527$, and $\delta_{\text{edge}} = 0.0121430664919214255\dots$, the exact coefficient is $c_{kL} = 7/2 + \delta_{\text{edge}}/6$. $\square$

Theorem 11.10 (Finite- L corrections to crossover moments). *The crossover moments are exactly 4, 12, 28 for all even $L > 4$. For $L = 4$, the first moment is 6, the second is 12, the third is 28.*

Proof. Direct computation of the Brillouin-zone sums for general even L . The identities $\sum \cos^2 = L/2$ and $\sum \cos^4 = 3L/8$ hold for $L > 4$; at $L = 4$, the first moment receives a correction of $+2$. $\square$

Theorem 11.11 (Floor function). *The floor function $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$ is the winding number of the DME-Core interface over one loop pass.*

Proof. The DME-Core interface phase is $\gamma_m(s) = e^{2\pi i \omega_m T_{\text{DME}} s}$, $s \in [0, 1)$. Its lift is $\widetilde{\gamma}_m(s) = \omega_m T_{\text{DME}} s$, which crosses integer values $\lfloor \omega_m T_{\text{DME}} \rfloor$ times per pass. Therefore $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$. $\square$

12 Philosophical and Physical Discussion

12.1 The status of time in HCSM

In HCSM, time is not fundamental. It is derived from the loop operator M_{loop} . The complex time $\tau_m = i \ln M_{\text{loop}}(m)$ is a readout of the loop's eigenvalue. The clock is the loop's phase $\theta_m = \arg M_{\text{loop}}(m)$; the arrow is the loop's dissipativity $|M_{\text{loop}}(m)| < 1$.

This is a radical departure from both general relativity and the Standard Model. In general relativity, time is a coordinate on a Lorentzian manifold. In the Standard Model, time is an external parameter in the Schrödinger equation and

the path integral. In HCSM, time is neither a coordinate nor a parameter; it is a derived quantity.

12.2 Comparison with general relativity

In general relativity, time is a coordinate t on a Lorentzian manifold. The metric $g_{\mu\nu}$ includes the time-time component g_{00} . The arrow of time is not explained; the Einstein equations are time-reversal symmetric. Time dilation arises from the metric. Proper time is $\int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$. There is no quantization of time.

In HCSM, time is derived from the loop operator. The arrow of time is derived from the loop's dissipativity. Time dilation arises from the phase handoff $\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}$. Proper time is the loop count k times T_{DME} . Time is quantized by the floor function $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.

12.3 Comparison with the Standard Model

The Standard Model has no fundamental arrow of time. The Schrödinger equation is time-reversal symmetric; CP violation is a small effect that does not explain the cosmological arrow. In HCSM, the arrow of time is derived from the loop's dissipativity. The Core gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$ is irreversible; the loop eigenvalues satisfy $|M_{\text{loop}}(m)| < 1$.

12.4 The unification of time and motion

In general relativity and the Standard Model, time and motion are separate concepts. In HCSM, they are the same object. Motion is the loop operator's composition; time is the loop operator's logarithm. They are related by

$$\boxed{\text{Time} = i \ln(\text{Motion})}. \quad (47)$$

This is not a metaphor. It is a structural identity. The loop operator M_{loop} is the fundamental object; time and motion are its two faces.

12.5 Dimensions are not fundamental

In general relativity and the Standard Model, spacetime dimension $d = 4$ is fundamental. In

HCSM, $d = 4$ is derived from the topology of the Hodge-Dirac kernel:

$$d = \dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (48)$$

Dimension is not the stage on which physics happens. It is a readout of the substrate's cohomology. Time and dimension are siblings: both are derived from the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

12.6 The substrate as the fundamental object

In HCSM, the only fundamental object is the substrate. Everything else is derived: dimension $d = 4$ from $\dim \ker D$; time from $\tau_m = i \ln M_{\text{loop}}(m)$; motion from $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$; spin from $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle \gamma_5, \chi_{\text{spin}}, F \rangle$; gauge algebra from the D_4 commutant; particle masses from the mass map; mixing matrices from the substrate.

The substrate is the machine. Time, motion, and dimension are its readouts.

12.7 Implications for quantum gravity

HCSM suggests a resolution of the problem of time in quantum gravity. In canonical quantum gravity, the Wheeler-DeWitt equation $\hat{H}\Psi = 0$ has no external time parameter. In HCSM, time is derived from the loop operator, not assumed. The loop operator is the fundamental object; time is its logarithm. This provides a natural relational notion of time without requiring an external clock.

12.8 Implications for cosmology

HCSM predicts a specific dark energy equation of state $w = -1.0140540788\dots$, testable by DESI DR3 and Euclid. It predicts $\theta_{\text{QCD}} = 0$ exactly, solving the strong CP problem. It predicts no light dark photon ($\epsilon \equiv 0$). These predictions distinguish HCSM from the Standard Model and general relativity.

13 Falsifiable Predictions

13.1 Internal consistency checks

- (1) τ_A and τ_B exactly as stated; a 10^{-10} deviation falsifies.

- (2) $T = \Pi_{\text{Re}(\tau) > 0}$: a mode with $\text{Re}(\tau) > 0$ outside $\text{im}(T)$ (or vice versa) falsifies.
- (3) Arrow direction: any loop mode with $|M_j| > 1$ falsifies.
- (4) Kernel freezing: any nonzero time evolution of $\{\text{dark}, W, Z, H\}$ falsifies.
- (5) Loop-pass duration: any experimental or structural result requiring the pass duration to be T_{beat} falsifies Theorem 8.14.

13.2 Cosmological predictions

- (6) $w = -1.0140540788\dots$; testable at $> 3\sigma$ by DESI DR3 + Euclid.
- (7) $\zeta H/\rho = 0.0046846929\dots$; testable by Euclid growth-rate and cosmic-shear analyses.
- (8) $\rho_{DE} = 2.5003 \times 10^{-47} \text{ GeV}^4$; matches observation to 0.010%.

13.3 Gravitational-wave predictions

- (9) $c_g = c_\gamma = c$ exactly. GW170817 constrains $|c_g/c_\gamma - 1| < 10^{-15}$.
- (10) No deviation from Newton's law at any observable scale.
- (11) $O(\epsilon^2)$ lattice corrections to the Newtonian potential are unobservably small.

13.4 Particle-physics predictions

- (12) No light dark photon: $\epsilon \equiv 0$ exactly.
- (13) Strong CP: $\theta_{\text{QCD}} = 0$ exactly.
- (14) ρ parameter: tree-level $\rho = 1$ exactly.
- (15) Fermion content: exactly 14 particles per generation.
- (16) $\sin \theta_{12} = \pi/14 = 0.2243995\dots$; matches observation to 0.044%.

14 Conclusion

We have derived time and its arrow in the Hodge Complex Standard Model as internal structures of the five-layer loop. The loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ acts diagonally on the 14-mode multiplet with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The complex time of mode m is $\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$.

The paper has established twelve principal results:

- (1) The prior, during, and after states at each of the five levels, with transition rules.
- (2) Complex time as the loop's logarithm: $\tau_m = i \ln M_{\text{loop}}(m)$.
- (3) Arrow of time from dissipativity: $|M_{\text{loop}}(m)| < 1$.
- (4) Topological floor function: $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.
- (5) Dimensional homogeneity of τ_m .
- (6) Exact ratio $|M_A|/|M_B| = 3 = d - 1$.
- (7) Landauer selection of T_{DME} .
- (8) Spin chirality $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ commutes with the loop.
- (9) Time unit $T_{\text{unit}} = 2\pi/m_s \approx 5.3$ days.
- (10) Particle identification from the Midpoint Sorting Theorem.
- (11) Continuous time $t = kT_{\text{DME}}$.
- (12) Philosophical and physical discussion comparing HCSM with GR and the SM.

The elegant theorem is stated: *Time is the loop's logarithm; motion is the loop's composition; the loop is the machine.*

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none. Physics is what the machine cannot cancel. The Hodge complex is the machine. Time is its logarithm. Motion is its composition.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-03: Lorentzian Signature from the Hodge Star*, HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-05: D_4 Representation Theory of the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-14: The Winding Generator on the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-20: The Motion Theorem*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-23 (Seventh Revision): The Midpoint Sorting Theorem*, HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-54: Vacuum Stability and the Stability Coefficient*, HCSM White Paper Series (2026).
- [12] S. Preschutti, *HCSM-56: Vacuum EFT*, HCSM White Paper Series (2026).
- [13] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM J. Res. Dev. **5**, 183 (1961).
- [14] J. Dodziuk, *Finite-difference approach to the Hodge theory of harmonic forms*, Amer. J. Math. **98**, 79 (1976).

- [15] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024).